**Code No: 115EM** **R13**

**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD**
**B. Tech III Year I Semester Examinations, May/June - 2019**
**SOFTWARE ENGINEERING**
**(Common to CSE, IT)**

**Time: 3 hours** **Max. Marks: 75**

**Note:** This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

**PART - A**

**(25 Marks)**

| | | |
|---|---|---|
| 1.a) | Define software engineering. | [2] |
| b) | List evolutionary process models. | [3] |
| c) | Differentiate between user requirement and system requirement. | [2] |
| d) | List the various types of feasibility studies. | [3] |
| e) | What are the goals of the design process? | [2] |
| f) | Define software architecture. | [3] |
| g) | What is meant by smoke testing? | [2] |
| h) | List the metrics for source code. | [3] |
| i) | Give the different categories of risks. | [2] |
| j) | What is meant by software review? | [3] |

**PART - B**

**(50 Marks)**

2. Discuss managers myths about software development and their effect on the practitioners performance as well as on overall outcome. [10]

**OR**

3. What is software process? What is need of software process improvement? Discuss capability maturity models. [10]

4. "The functional requirements specification of a system should be both complete and consistent". Substantiate this statement with relevant examples. [10]

**OR**

5.a) Draw a context level model for a web-based food- ordering system such as "Swiggy".
b) Discuss the main characteristics of data model for requirement engineering. [5+5]

6. How to translate the analysis model into the design model? Explain with an example scenario. [10]

**OR**

7.a) Explain how to map data flow into a software architecture?
b) Explain the design of class based components. [5+5]

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8.a) Discuss the process of debugging.
b) What is the need of beta testing? [5+5]

**OR**

9.a) Explain the metrics for software quality.
b) Explain about the test strategies for connectional software. [5+5]

10.a) Elaborate on risk projection steps.
b) Provide the format of risk information sheet. [5+5]

**OR**

11. Explain the activities of software quality assurance group to assist the software team in achieving high quality. [10]

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